Assignment 3

Summary

This assignment involves exploring core object-oriented programming concepts by creating and using objects that represent shapes.

Submission Checklist

Submit the following in Slate:

The modified C++ source file named **`assign3.cpp`**

Requirements

Program Code

Your code **must** meet all the following conditions:

- Must have no syntax errors in your submission.
- Can only include standard libraries discussed in class.
- **Use only techniques and concepts demonstrated in the course in addition to any specific instructions provided in this assignment.**
- **Use best practices shared in class.** Code should be clear, concise, properly organized into appropriately named files and adhere to standards and naming conventions discussed during class.
- Where prompted, assume the user enters data of the correct **type** where input is requested.

In your code, you **must provide meaningful and relevant comments and structure** that aligns with C/C++ industry standards. For this assignment, this must include:

1. Comments that describe the purpose of variables, branching, looping, and functions (with the definition, not the prototype).
2. Use of variable and function names that are descriptive and use camelCase. No abbreviations (e.g.: `x, n, i, f, ch`) except when part of a C idiom.
3. The use of function prototypes with your code structured as:

Signature and description
Libraries and constants (if any)
Function prototypes
Class declarations and definitions
Main function
Function definitions

4. Implement an object-oriented design using classes and objects in C++

Academic Integrity

1. This assignment shall be completed **individually**. All work must be your own. Copying or reproducing the work done by others (in part or in full) or letting others to copy or reproduce your own work is subject to significant grade reduction or getting no grade at all and/or being treated as academic dishonesty under the College's Academic Dishonesty Policy.
2. Use of AI to write and/or submit code, in part or in whole, is not allowed.
3. **Reminder: if you want to include code found elsewhere, or not sure about how you can use a resource or technique, please consult with your professor.**

Submitting the Assignment

1. The assignment shall be submitted by the specified due date. Late submissions will be penalized with 10% per day for up to 3 calendar days after which the assignment cannot be submitted anymore. Assignments are not accepted after the final deadline.
2. Submitting the assignment is done using the SLATE Assignment folder. DO NOT email your submission.

Programs

Complete the code in the provided C++ file. Instructions and requirements are included in the comments.